

Instructions for preparing the solution script:

- Write your name, ID#, and Section number clearly in the very front page.
 - Your submission must be a handwritten version.
 - Start working right away. No late submission will be accepted for the assignment.
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Assignment Task Description [10 Marks]:

You have to study the canonical LR(1), that is, CLR(1), parser construction from Section 4.7.1 to 4.7.3 of the reference [textbook](#).

To complete the assignment task, you need to **construct the LR(1) Automaton** for the following grammar. Also **build the CLR(1) Parsing Table** based on the collection of sets of LR(1) items $C' = \{I_0, I_1, I_2, I_3, \dots, I_n\}$ for the augmented grammar G' . Lastly, **simulate the parsing process** using a stack for the following input string.

Grammar:

$$V \rightarrow E$$
$$E \rightarrow FT \mid G$$
$$F \rightarrow float$$
$$T \rightarrow id$$
$$G \rightarrow \epsilon$$

Input String: *float id*
